Practical No . 4

Aim: Configure routers in a network using Static routing protocol in Cisco Packet Tracer and show transmission of packet from one network to another.

Theory:

Introduction

Static routing is a routing technique in which routes are **manually configured** by a network administrator in a router's routing table. It is one of the simplest routing methods used in computer networks and is commonly implemented in **small or stable networks** where the network topology does not change frequently.

In networking simulators like **Cisco Packet Tracer**, static routing helps in understanding how data packets are forwarded from one network to another through routers.

Static Routing

Static routing involves **manually specifying the path** that data packets must follow to reach a destination network. Unlike dynamic routing protocols, static routing does not automatically learn or update routes. Each route must be configured explicitly by the administrator. sOnce configured, the router uses these predefined routes to forward packets to the correct next-hop router or destination network.

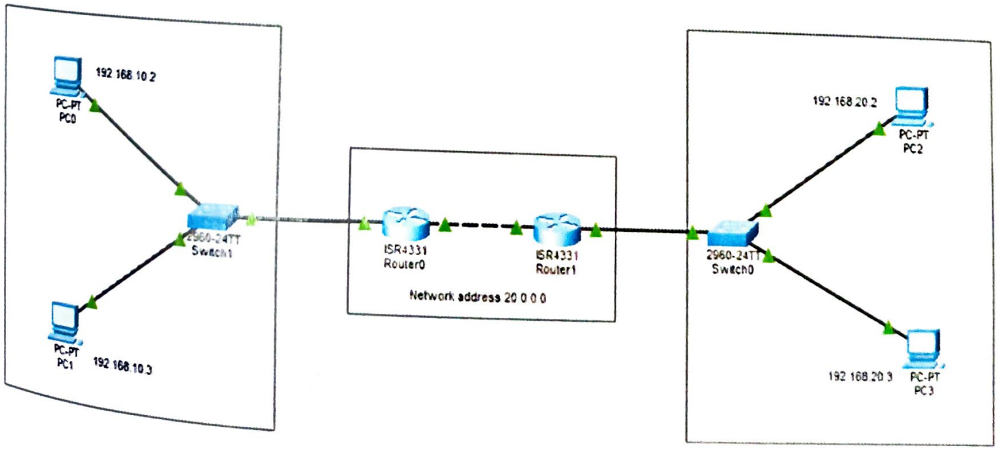
Working Principle of Static Routing

When a data packet is sent from a source network to a destination network:

1. The packet reaches the **default gateway router**, The router checks its **routing table** for a matching destination network.
2. If a static route exists, the router forwards the packet to the **next-hop router**. The process continues until the packet reaches the destination network.
3. The destination host receives the packet and sends a response back using the same routing logic.

Thus, static routing enables successful communication between **different networks connected through routers**.

Packet Transmission Between Networks



Packet transmission in static routing follows a **predefined path**. The router does not make dynamic decisions; instead, it forwards packets based solely on the static entries in its routing table. This ensures predictable routing behavior and controlled network traffic flow.

Steps to Configure Static Routing

Step 1: Create Network Topology

- Place **2 Routers, 2 Switches,** and **4 PCs** in Cisco Packet Tracer
- Connect:
  - PCs → Switches (Straight-through cable)
  - Switches → Routers
  - Router0 ↔ Router1 (Serial / Gigabit link)

Step 2: Assign IP Addresses

- Left LAN: 192.168.10.0/24
- Right LAN: 192.168.20.0/24
- Router-to-router network: 20.0.0.0

Assign IPs to PCs and router interfaces as shown in the diagram.

Step 3: Configure Router Interfaces

- Enable router interfaces
- Assign IP address to:
  - LAN interface
  - Inter-router interface
- Use **no shutdown** on all interfaces

Step 4: Configure Static Routes

- On **Router0**, add route for 192.168.20.0 network via Router1
- On **Router1**, add route for 192.168.10.0 network via Router0

(This allows communication between both LANs.)

Step 5: Configure Default Gateway on PCs

- Left PCs → Gateway: Router0 LAN IP
- Right PCs → Gateway: Router1 LAN IP

Step 6: Verify Packet Transmission

- Use **ping** from one LAN PC to another LAN PC
- Switch to **Simulation Mode** to observe ICMP packet flow

Advantages of Static Routing

- Simple and easy to configure
- Requires **no routing protocol overhead**
- More secure since routes are not advertised
- Suitable for **small networks**

Disadvantages of Static Routing

- Not suitable for large or complex networks
- Requires manual reconfiguration if the network changes
- No automatic failure detection or rerouting

Applications of Static Routing

- Small enterprise networks
- Branch office connections
- Backup routes in combination with dynamic routing
- Lab environments and network learning simulations

Conclusion

Static routing provides a **basic yet effective method** for connecting multiple networks through routers. By manu defining routes, routers can successfully forward packets from one network to another. Although static routing la scalability and adaptability, it plays a crucial role in **network fundamentals and learning environments** such a Cisco Packet Tracer.

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